1. Define Data Mining and list any two basic data mining tasks. Explain the steps Involved in KDD process and justify the role of data preprocessing in it

• Data mining is the process of discovering useful pattern, relationships and knowledge from large amount of data

Two Basic Data Mining Tasks

1. classification
2. clustering

2. what is Data cleaning mention two common data quality issues. mention two common methods to handle missing values and Explain noisy data

• Data cleaning is the process of detecting and correction errors, missing values, noisly data, and duplicate records

Data Quality Issues

1. Missing values
2. Noisly data

Age values: 20, 21, 22, ?, 23

with average = 21.5 ≃ 22

3. Define Data Integration and list two challenges
Explain redundancy and Inconsistency

• Data Integration combines data from multiple source into one unified dataset

challenges:-

1. Data redundancy
2. Data inconsistency

4. What is convariance Analysis? List the two convariance formulas

convariance measure how two variables change together

Formula (population)

$$cov(x,y) = \frac{\Sigma(x-x)(y \cdot y)}{n}$$

Formula (Sample)

$$cov(x,y) = \frac{\Sigma(x-x)(y-y)}{n-1}$$

5. Suppose two stocks A and B have the following values in one week (2,5)(3,8) (5,10).(4,11).

(2,5) (3,8) (5,10,11,14)

Covariance is positive, so both stocks rise together

They are not Independent because they move in the same direction

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Explain the architecture of a typical data mining system

components

- Data sources
- Database
- Data cleaning
- Data mining engine
- pattern Evaluation
- Knowledge Base
- User interface

7. Find covariance

Given: $X = \{4, 5, 6, 7, 9\}$ $Y = \{8, 3, 7, 5, 6\}$

Mean of $X = 6.2$

Mean of $Y = 5.8$

Sample covariance $= -0.25$

8. Describe the major step in data preprocessing

- Data cleaning
- Data Integration
- Data Transformation
- Data Reduction
- Data Discretization

9. Equal Frequency (equi-depth) Binning

Data: 13, 15, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70

• There are 26 values

Using 5 equal frequency bins:

Bin1: 13, 15, 16, 19, 20

Bin2: 20, 21, 22, 22, 25

Bin3: 25, 25, 25, 30, 33

Bin4: 33, 35, 35, 35, 35

Bin5: 36, 40, 45, 46, 52, 70

10 Explain data discretization and compare equal width and equal frequency binning

Data discretization converts continuous values into intervals

Importance:

- Reduces complexity
- Improves mining efficiency
- Makes data easier to understand.